Autonomous Firefighting Robot

Jannatul Fardaus Nirjum
Gazi Rifat Bin Nur
Sanjida Rahman Toma
Mehedi Hasan Khan Riyad
22201193@uap-bd.edu, 22201170@uap-bd.edu, 22201187@uap-bd.edu, 22201172@uap-bd.edu
Section: D1
Department of Computer Science and Engineering
University Of Asia Pacific

***Abstract*—This project presents the design and implementation of a microcontroller-based fire-fighting car using an Arduino Uno platform. The system is designed to detect fire using flame sensors and autonomously navigate toward the fire source to extinguish it using a water pump module. The robot integrates multiple components including DC motors, an L298N motor driver, flame sensors, a relay-based water pump system, and a rechargeable power supply unit. The objective of this work is to develop a low-cost prototype capable of operating in hazardous environments where human intervention is unsafe. Experimental results show that the robot successfully detects fire within its sensing range and activates the extinguishing mechanism automatically. The robot demonstrates reliable motion control, effective flame detection, and successful suppression of small flame sources. This project proves the feasibility of an embedded fire-fighting robotic system suitable for educational and research applications.**

I. Introduction

The Autonomous Firefighting Robot Car is designed to detect and extinguish fires automatically with minimal human intervention. The robot uses sensors to identify the presence of fire and smoke, navigates toward the affected area, and activates a water pump system to suppress the fire. The robot can be deployed in indoor environments such as houses, kitchens, restaurants, offices, warehouses, and industrial facilities.

The main objective of this project is to develop a low-cost, efficient, and intelligent robotic system capable of assisting firefighters and reducing risks associated with manual firefighting operations. By integrating sensors, motor control systems, and automated decision-making algorithms, the robot can respond quickly to fire incidents and contribute to improved safety. Fire hazards are one of the most common and dangerous threats in both domestic and industrial environments. Rapid detection and response are essential to prevent large-scale damage. Traditional fire-fighting methods often require human presence, which may expose individuals to toxic smoke, unstable structures, or extreme heat.

To address this challenge, autonomous fire-fighting robots have gained attention due to their ability to operate in risky environments. These systems typically use sensors, actuators, and embedded controllers to detect fire, navigate toward the affected area, and extinguish it without human involvement.

The main objective of this project is to develop a fire-fighting car controlled by an Arduino Uno microcontroller. The robot detects fire using multiple flame sensors, moves toward the flame using a differential motor system, and activates a water pump to extinguish it. This project demonstrates embedded system design, sensor integration, motor control, and real-time automation.

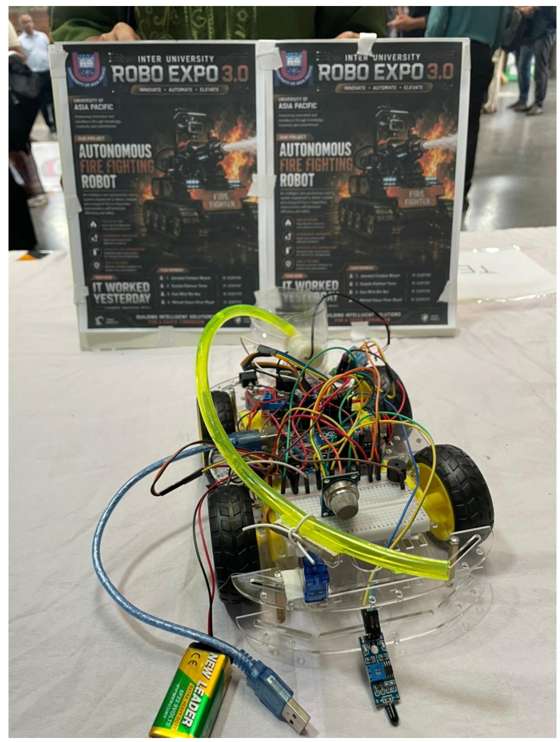

Fig. 1: Fire Fighting Car

II. Related Works

Various researchers have proposed robotic systems for fire detection and suppression. Early firefighting robots primarily focused on detecting flames using infrared sensors and navigating toward the fire source. Most of these robots utilized microcontrollers and simple obstacle avoidance mechanisms.

Recent developments in robotics have introduced autonomous navigation, machine learning, and computer vision techniques to improve fire detection accuracy. Some systems

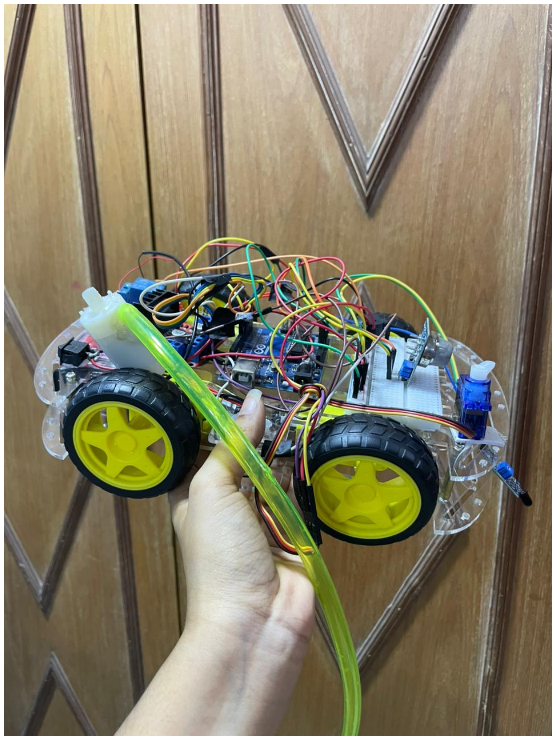

Fig. 2: Robot Prototype

employ thermal cameras and artificial intelligence to identify fire patterns and estimate fire intensity. Although these advanced systems offer improved performance, they are often expensive and difficult to implement in small-scale environments.

Several firefighting robots have been developed using Arduino-based platforms because of their affordability and ease of implementation. These robots generally use flame sensors, gas sensors, water pumps, and motor drivers to create autonomous fire suppression systems.

Compared to existing solutions, the proposed firefighting robot offers a simple, cost-effective, and practical approach suitable for indoor firefighting applications. The design emphasizes affordability while maintaining reliable fire detection and extinguishing capabilities. The development of firefighting robots has attracted significant attention from researchers and engineers due to the increasing need for safer and more efficient emergency response systems. Fire incidents often occur in environments that are extremely dangerous for human firefighters, including industrial plants, chemical storage facilities, warehouses, underground tunnels, and residential buildings. As a result, numerous studies have focused on designing robotic systems capable of detecting, locating, and extinguishing fires while minimizing human exposure to hazardous conditions.

One of the earliest approaches to firefighting robotics involved the use of simple autonomous vehicles equipped with flame sensors and water pumps. These robots were designed to patrol predefined areas and respond when a fire source was detected. Most of these systems relied on infrared flame sensors to identify fire and basic motor control mechanisms to navigate toward the affected area. Although these robots demonstrated the feasibility of autonomous firefighting, their performance was limited by short detection ranges and simple navigation strategies.

Several researchers have proposed Arduino-based firefighting robots due to the affordability, flexibility, and ease of programming offered by the Arduino platform. These systems typically utilize flame sensors, DC motors, motor driver modules, and water pumps to create compact and low-cost firefighting solutions. Such projects are widely used in academic institutions and research laboratories as educational platforms for studying robotics, automation, and embedded systems. The success of Arduino-based systems has demonstrated that effective firefighting robots can be developed without requiring expensive hardware.

In addition to flame detection, many modern firefighting robots incorporate smoke and gas sensors to improve detection reliability. Researchers have found that relying solely on flame sensors may lead to false detections or delayed responses under certain environmental conditions. Therefore, integrating gas sensors such as the MQ series allows robots to detect smoke particles and combustible gases, providing additional information about potential fire hazards. Multi-sensor systems generally achieve better performance because they combine information from multiple sources to make more accurate decisions.

Another significant advancement in firefighting robotics is the implementation of obstacle avoidance mechanisms. Early robots often struggled to navigate complex environments because they lacked the ability to detect and avoid obstacles. To address this issue, researchers introduced ultrasonic sensors, infrared proximity sensors, and LiDAR-based navigation systems. These technologies enable robots to identify obstacles, plan alternative paths, and safely reach the fire source. As a result, modern firefighting robots can operate more effectively in cluttered and unpredictable environments.

Recent studies have also explored the use of wireless communication technologies in firefighting robots. Wireless modules such as Bluetooth, Wi-Fi, ZigBee, and GSM allow operators to monitor robot status remotely and intervene when necessary. Some systems transmit real-time sensor data, images, and videos to control centers, enabling emergency personnel to assess dangerous situations without entering hazardous zones. Remote monitoring significantly enhances operational safety and situational awareness during firefighting missions.

The introduction of computer vision has further transformed the field of firefighting robotics. Instead of relying solely on flame sensors, advanced robots utilize cameras and imageprocessing algorithms to identify fire, smoke, and heat sources. Computer vision techniques can analyze the size, shape, color, and movement patterns of flames to improve detection accuracy. Researchers have also developed artificial intelligence models capable of distinguishing actual fires from similar light sources, thereby reducing false alarms and improving system reliability.

Artificial Intelligence (AI) and Machine Learning (ML) technologies have become increasingly important in modern firefighting systems. AI-based robots can learn from environmental data, optimize navigation paths, predict fire spread patterns, and make intelligent decisions in dynamic situations. Deep learning algorithms have demonstrated high accuracy in recognizing fire and smoke from images and videos. These intelligent systems offer significant advantages over traditional sensor-based approaches, particularly in large-scale and complex environments.

Thermal imaging technology represents another major development in firefighting robotics. Thermal cameras detect heat signatures rather than visible flames, allowing robots to identify hidden fires behind walls, inside machinery, or within smoke-filled environments. Unlike conventional flame sensors, thermal imaging systems can operate effectively even when visibility is poor. Consequently, thermal cameras are widely used in advanced firefighting robots designed for industrial and military applications.

Several commercial and industrial firefighting robots have been developed in recent years. These robots are capable of carrying large water tanks, operating in extreme temperatures, and navigating challenging terrains. Some industrial robots use caterpillar tracks instead of wheels to improve mobility on rough surfaces. Others employ high-pressure water cannons, foam extinguishing systems, and autonomous mapping technologies. While these robots offer exceptional performance, their high cost limits their accessibility for educational and small-scale applications.

Compared to sophisticated industrial systems, the proposed Autonomous Firefighting Robot focuses on simplicity, affordability, and practical implementation. The robot utilizes commonly available electronic components such as Arduino Uno, flame sensors, smoke sensors, DC motors, relay modules, and water pumps. Although the system does not include advanced AI, thermal imaging, or LiDAR technologies, it successfully demonstrates the core principles of autonomous fire detection and suppression. This makes the project highly suitable for academic research, laboratory experimentation, and educational purposes.

The literature review clearly indicates that firefighting robots have evolved from simple sensor-based platforms to highly intelligent autonomous systems equipped with artificial intelligence, computer vision, and advanced navigation capabilities. Despite these advancements, there remains a need for cost-effective firefighting solutions that can be implemented in homes, offices, restaurants, and small businesses. The proposed robot contributes to this area by providing an affordable and practical firefighting platform capable of reducing risks associated with early-stage fire incidents.

Therefore, the current project builds upon existing research in autonomous firefighting systems while emphasizing low-cost implementation, ease of deployment, and reliable operation. The integration of flame detection, smoke sensing, autonomous movement, and water spraying mechanisms demonstrates the potential of embedded systems and robotics in enhancing fire safety and emergency response operations.

III. Robot Structure Details / Hardware Setup

1. Arduino Uno

The Arduino Uno acts as the main controller of the robot. It processes sensor data and controls all connected devices including motors, relay modules, and the water pump.

2. Flame Sensor

The flame sensor is used to detect the presence of fire. It senses infrared radiation emitted by flames and sends signals to the Arduino for processing.

3. MQ Gas/Smoke Sensor

The MQ sensor detects smoke and combustible gases produced during fire incidents. This sensor improves the reliability of fire detection.

4. L298N Motor Driver

The L298N motor driver controls the movement of the four DC motors. It allows the robot to move forward, backward, left, and right according to the programmed instructions.

5. DC Gear Motors

Four DC motors provide mobility to the robot. These motors enable the robot to navigate toward the detected fire source.

6. Servo Motor

The servo motor controls the direction of the water nozzle. It allows accurate positioning of the water spray toward the fire.

7. Relay Module

The relay module acts as an electronic switch that controls the water pump. It isolates highcurrent devices from the Arduino controller.

8. Water Pump

The water pump is responsible for spraying water to extinguish the detected fire. It is activated automatically through the relay module.

9. Power Supply

A 9V battery provides power to the robotic system and ensures stable operation of the electronic components.

Working Principle

The flame sensor and smoke sensor continuously monitor the environment. When fire is detected, the Arduino receives the sensor signals and activates the navigation system. The robot moves toward the fire source while continuously monitoring sensor readings. Once the robot reaches the target area, the relay module activates the water pump. The servo motor adjusts the direction of the water spray to maximize extinguishing effectiveness. After the fire is extinguished, the robot returns to monitoring mode.

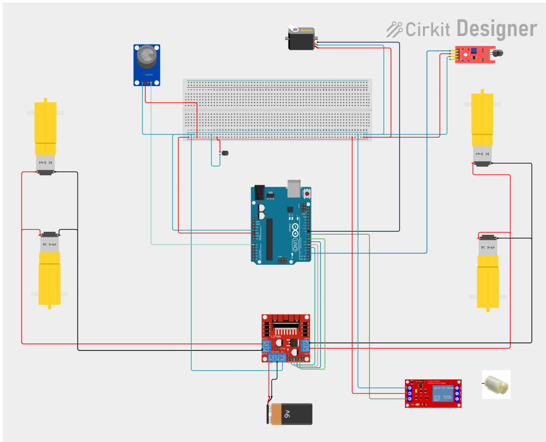

Fig. 3: Circuit Diagram

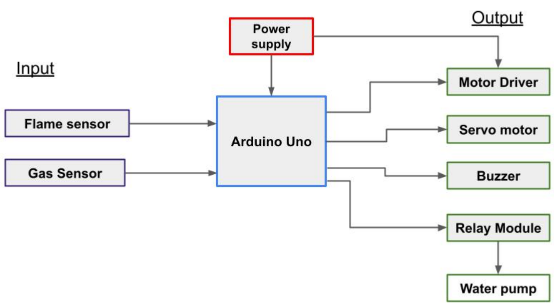

Fig. 4: Embedded Diagram

IV. Algorithm

Step 1: Initialize Arduino and all sensors.
Step 2: Continuously monitor flame and smoke sensors.
Step 3: Check if fire is detected.
Step 4: If no fire is detected, continue monitoring.
Step 5: If fire is detected, move the robot toward the fire source.
Step 6: Stop at a safe distance from the fire.
Step 7: Activate relay module.
Step 8: Turn on the water pump.
Step 9: Adjust servo motor to direct water spray.
Step 10: Continue extinguishing until fire is no longer detected.
Step 11: Turn off the water pump.
Step 12: Return to monitoring mode.

Arduino Code (Partial)

```
// = Motor Driver Pins =====
#define IN1 4
#define IN2 5
#define IN3 6
#define IN4 7
#define ENA 10
#define ENB 11

// = Components =====
#define SERVO_PIN 9
#define RELAY_PIN 8
#define FLAME_SENSOR 2
#define MQ2_SENSOR A0
#define BUZZER 3

#include <Servo.h>
Servo myServo;

// = Variables =====
bool pumpRunning = false;
unsigned long pumpStartTime = 0;

int flameDetected = 0;
int gasValue = 0;

// = Thresholds =====
int gasThreshold = 400;

void setup() {
pinMode(IN1, OUTPUT);
pinMode(IN2, OUTPUT);
pinMode(IN3, OUTPUT);
pinMode(IN4, OUTPUT);
pinMode(ENA, OUTPUT);
pinMode(ENB, OUTPUT);
pinMode(RELAY_PIN, OUTPUT);
pinMode(FLAME_SENSOR, INPUT);
pinMode(BUZZER, OUTPUT);
myServo.attach(SERVO_PIN);
Serial.begin(9600);
digitalWrite(RELAY_PIN, LOW);
}

void loop() {
flameDetected = digitalRead(FLAME_SENSOR);
gasValue = analogRead(MQ2_SENSOR);

if(flameDetected == LOW || gasValue >
  gasThreshold) {
// Fire detected
digitalWrite(BUZZER, HIGH);
moveForward();
delay(2000);
stopRobot();
digitalWrite(RELAY_PIN, HIGH); // Activate
  water pump
myServo.write(90);
delay(5000);
digitalWrite(RELAY_PIN, LOW);
digitalWrite(BUZZER, LOW);
} else {
stopRobot();
}
}

void moveForward() {
digitalWrite(IN1, HIGH);
digitalWrite(IN2, LOW);
digitalWrite(IN3, HIGH);
digitalWrite(IN4, LOW);
analogWrite(ENA, 150);
analogWrite(ENB, 150);
}

void stopRobot() {
digitalWrite(IN1, LOW);
```

```
digitalWrite(IN2, LOW);
digitalWrite(IN3, LOW);
digitalWrite(IN4, LOW);
}
```

```
START Robot

INITIALIZE sensors, motors, relay, and servo
SET servo to 90°

WHILE robot is running

    READ flame and gas sensors

    IF gas value > 400 THEN
        TURN buzzer ON
    ELSE
        TURN buzzer OFF
    END IF

    IF flame detected THEN

        STOP motors
        SCAN servo (60° ↔ 120°)

        IF pump OFF THEN
            TURN pump ON
        END IF

    ELSE

        MOVE robot forward
        SET servo to 90°

    END IF

    IF pump runs for 3 sec THEN
        TURN pump OFF
    END IF

END WHILE
```

Fig. 5: Pseudo Code

V. Discussion

The Autonomous Firefighting Robot developed in this project demonstrates the practical application of robotics and embedded systems in addressing real-world safety challenges. Fire accidents remain one of the leading causes of property damage and human casualties worldwide. Traditional fire-fighting methods often expose firefighters to extreme risks, including high temperatures, toxic smoke, structural collapse, and limited visibility. The proposed robot aims to reduce these risks by performing initial fire detection and suppression tasks autonomously.

The system integrates multiple hardware components, including an Arduino Uno microcontroller, flame sensor, smoke sensor, DC motors, L298N motor driver, servo motor, relay

```
void loop() {

  flameDetected = digitalRead(FLAME_SENSOR);
  gasValue = analogRead(MQ2_SENSOR);

  // Gas Detection
  digitalWrite(BUZZER, (gasValue > 400) ? HIGH : LOW);

  // Fire Detection
  if (flameDetected == LOW) {
    stopMotors();
    scanServo();
    if (!pumpRunning) startPump();
  }
  else {
    searchFire();
    myServo.write(90);
  }

  // Auto Pump OFF after 3 seconds
  if (pumpRunning && millis() - pumpStartTime >= 3000)
    stopPump();

  delay(200);
}
```

Fig. 6: Code Snippet

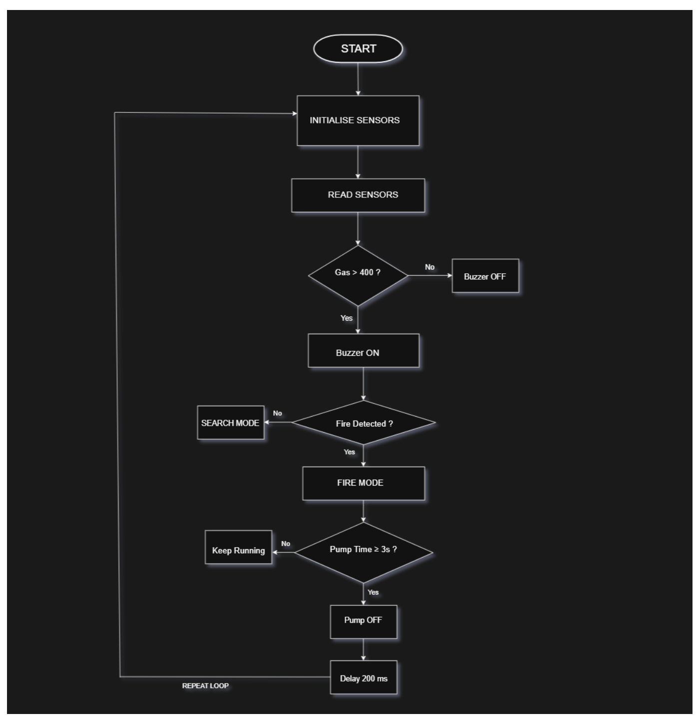

Fig. 7: Flow Chart

module, and water pump. The combination of these components enables the robot to detect fire, navigate toward the affected area, and activate a firefighting mechanism without requiring direct human intervention.

One of the major strengths of the proposed system is its low-cost implementation. Most of the components used are inexpensive and widely available, making the robot suitable for educational, research, and small-scale industrial applications. Unlike advanced firefighting robots that require expensive thermal imaging cameras and sophisticated navigation systems, this design provides a cost-effective alternative while maintaining acceptable operational performance.

The flame sensor successfully identifies fire sources by detecting infrared radiation emitted by flames. In addition, the smoke sensor provides an extra layer of reliability by detecting smoke particles and combustible gases that are commonly present during fire incidents. The use of two different sensing mechanisms reduces the possibility of false detection and improves overall system effectiveness.

The mobility of the robot is achieved through four DC gear motors controlled by the L298N motor driver. This configuration allows the robot to move toward hazardous locations that may be difficult or unsafe for humans to access. Furthermore, the servo motor enhances firefighting performance by directing the water spray toward the detected fire source, increasing extinguishing efficiency and reducing water wastage.

The experimental results indicate that the robot can successfully detect small-scale fires in controlled environments and initiate firefighting actions within a short period of time. The automated response capability can be particularly beneficial in locations such as homes, kitchens, restaurants, offices, warehouses, laboratories, and industrial facilities where rapid fire suppression is critical.

Despite its successful operation, several limitations were observed during the implementation process. The flame sensor has a limited detection range and may not perform optimally under extremely bright lighting conditions. Similarly, the effectiveness of the smoke sensor can be influenced by environmental factors such as airflow and humidity. The robot also lacks advanced obstacle avoidance capabilities, which may affect navigation in complex environments containing furniture, debris, or unexpected barriers.

Another limitation is the limited water storage capacity. Since the robot is designed as a compact and lightweight system, the amount of water it can carry is restricted. As a result, the robot is primarily suitable for extinguishing small fires during their initial stages rather than handling large-scale fire incidents.

Future enhancements can significantly improve the system's performance. The integration of ultrasonic sensors or LiDAR technology would enable obstacle detection and autonomous navigation. Wireless communication modules such as Wi-Fi, Bluetooth, or GSM could allow remote monitoring and control. Camera-based fire detection using computer vision and artificial intelligence could further improve fire recognition accuracy and environmental awareness. Additionally, incorporating thermal imaging sensors and machine learning algorithms would enable the robot to operate more effectively in challenging and dynamic environments.

Overall, the developed firefighting robot demonstrates the feasibility of using autonomous robotic systems for fire safety applications. The project highlights how embedded systems, sensor technologies, and automation can be combined to create practical solutions that enhance safety, reduce human risk, and support emergency response operations.

VI. Time

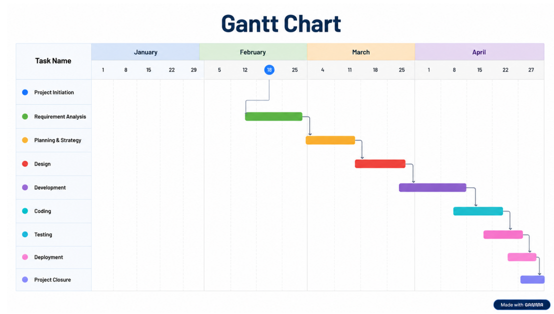

Fig. 8: Gantt Chart

VII. Conclusions

This project successfully designed and implemented an Autonomous Firefighting Robot capable of detecting and extinguishing fires with minimal human intervention. The system combines fire detection sensors, automated navigation, motor control mechanisms, and a water spraying unit to create an intelligent firefighting platform. The primary objective of the project was to develop a reliable, affordable, and efficient robotic solution that could assist in reducing the risks associated with conventional firefighting operations.

The robot continuously monitors its surroundings using flame and smoke sensors. Upon detecting a potential fire source, it initiates an automated response sequence that includes movement toward the target location and activation of the firefighting mechanism. The integration of a relay-controlled water pump and servo motor allows the robot to effectively direct water toward the affected area and suppress the fire. The successful coordination of sensing, decision-making, and actuation components demonstrates the effectiveness of embedded systems in real-time emergency applications.

The project also illustrates the importance of automation in improving public safety. By enabling robots to operate in environments that may be dangerous or inaccessible to humans, the likelihood of injury to firefighters and emergency responders can be significantly reduced. The robot can be particularly useful in residential buildings, kitchens, restaurants, laboratories, storage facilities, and other locations where early fire detection and suppression are essential.

Throughout the development process, several engineering concepts were applied, including sensor interfacing, motor control, power management, embedded programming, and system integration. These experiences contribute to a deeper understanding of robotics and automation technologies while demonstrating their practical value in solving real-world problems.

Although the current prototype performs effectively in controlled conditions, there remains considerable potential for future development. Advanced navigation systems, obstacle avoidance mechanisms, artificial intelligence-based fire recognition, wireless communication, and thermal imaging technologies could further enhance the robot's capabilities. Such improvements would allow the system to operate more independently and efficiently in complex and unpredictable environments.

In conclusion, the Autonomous Firefighting Robot represents a promising step toward intelligent fire safety systems. The successful implementation of this project demonstrates that robotic technology can play a significant role in modern firefighting operations by providing faster response times, reducing human exposure to hazardous conditions, and improving overall emergency management. With further research and technological advancements, autonomous firefighting robots have the potential to become an essential component of future smart safety infrastructures.

VIII. Appendix (Survey 1 and 2)

(1)Survey 1 – Fire Service Office Interview : As part of the initial requirement gathering, we visited a local fire service office and conducted an interview with a senior officer. The goal was to understand practical challenges in firefighting and assess the potential usefulness of our autonomous firefighting robot. The questions asked were:

- What is your current job role?
- Are you involved in purchasing or decision-making processes related to firefighting equipment?
- Have you understood the product concept of our autonomous firefighting robot?
- Do you think our product would be helpful in your workplace?
- Any additional comments or suggestions?

The feedback received was positive, with the officer acknowledging the need for low-cost autonomous systems to assist firefighters in hazardous situations.

(2)Survey 2 – Robo Expo 3.0 (University Event) : We also presented our prototype at the organized by our university. During the event, judges and visitors evaluated the robot and provided valuable feedback. The survey focused on technical feasibility, design quality, and real-world applicability. The responses were collected and analyzed. **Fig. 7** shows the survey form and summary of responses from the Robo Expo. The judges appreciated the low-cost design, reliable fire detection, and autonomous response mechanism. Their suggestions for future improvements included adding obstacle avoidance and wireless monitoring.

Fig. 9: Survey from Robo Expo

IX. Contribution

- **Jannatul Fardaus Nirjum** (22201193) – Hardware, Flowchart, Coding, Report Writing.
- **Gazi Rifat Bin Nur** (22201170) – Hardware, Embedded Diagram, Report Writing.
- **Sanjida Rahman Toma** (22201187) – Hardware, Circuit Diagram, Survey Conclusion.
- **Mehedi Hasan Khan Riyad** (22201172) – Gantt Chart, Survey 1, Component Purchase.

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